

Customer Shopping Behavior Analysis

1. Project Overview

This project focuses on analyzing customer shopping behavior using transactional data containing **3,900 purchase records** across multiple product categories. The primary objective is to understand customer spending habits, purchasing preferences, customer segments, and subscription patterns.

The analysis aims to generate meaningful business insights that can help organizations improve sales performance, strengthen customer relationships, and make more informed strategic decisions.

2. Dataset Summary

- **Number of Rows:** 3,900
- **Number of Columns:** 18

Key Features

Customer Demographics

- Age
- Gender
- Location
- Subscription Status

Purchase Information

- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Color

Customer Shopping Behavior

- Discount Applied
- Promo Code Used
- Previous Purchases

- Purchase Frequency
- Review Rating
- Shipping Type

Missing Data

The dataset contains **37 missing values in the Review Rating column**. These missing values were addressed during the data-cleaning stage.

3. Exploratory Data Analysis Using Python

Python was used for data preparation, cleaning, exploration, and feature engineering.

Data Loading

The dataset was imported into Python using the **Pandas** library for further analysis.

Initial Data Exploration

The structure and characteristics of the dataset were examined using functions such as:

- `df.info()` to inspect columns, data types, and missing values

#	Column	Non-Null Count	Dtype
0	Customer ID	3900 non-null	int64
1	Age	3900 non-null	int64
2	Gender	3900 non-null	object
3	Item Purchased	3900 non-null	object
4	Category	3900 non-null	object
5	Purchase Amount (USD)	3900 non-null	int64
6	Location	3900 non-null	object
7	Size	3900 non-null	object
8	Color	3900 non-null	object
9	Season	3900 non-null	object
10	Review Rating	3863 non-null	float64
11	Subscription Status	3900 non-null	object
12	Shipping Type	3900 non-null	object
13	Discount Applied	3900 non-null	object
14	Promo Code Used	3900 non-null	object
15	Previous Purchases	3900 non-null	int64
16	Payment Method	3900 non-null	object
17	Frequency of Purchases	3900 non-null	object

dtypes: float64(1), int64(4), object(13)

- `df.describe()` to obtain statistical summaries of numerical variables

	customer_id	age	purchase_amount	review_rating	previous_purchases
count	3900.000000	3900.000000	3900.000000	3900.000000	3900.000000
mean	1950.500000	44.068462	59.764359	3.750051	25.351538
std	1125.977353	15.207589	23.685392	0.713590	14.447125
min	1.000000	18.000000	20.000000	2.500000	1.000000
25%	975.750000	31.000000	39.000000	3.100000	13.000000
50%	1950.500000	44.000000	60.000000	3.800000	25.000000
75%	2925.250000	57.000000	81.000000	4.400000	38.000000
max	3900.000000	70.000000	100.000000	5.000000	50.000000

Handling Missing Values

Missing values in the **Review Rating** column were identified and replaced using the **median review rating for the respective product category**. This approach helped retain the available records while maintaining category-level consistency.

Column Standardization

Column names were converted into **snake_case format** to make them easier to work with in Python and SQL.

Feature Engineering

Additional features were created to support deeper analysis:

- **age_group** — Customers were grouped into meaningful age categories.
- **purchase_frequency_days** — Purchase-frequency information was converted into a numerical representation for analysis.

Data Consistency Check

The relationship between `discount_applied` and `promo_code_used` was examined to determine whether both variables provided unique information. Since the two fields were found to be redundant for the analysis, `promo_code_used` was removed.

Database Integration

After completing the cleaning and transformation process, the prepared DataFrame was connected to **MySQL** and uploaded to the database for further SQL-based analysis.

4. Data Analysis Using SQL

SQL was used to perform structured business analysis and answer important questions related to customer behavior, revenue, product performance, and customer loyalty.

1. Revenue Analysis by Gender

Total revenue was calculated separately for male and female customers to compare their contribution to overall sales.

	gender	total_revenue
▶	Male	157890
	Female	75191

2. High-Spending Customers Using Discounts

Customers who received discounts were analyzed to identify those whose purchase amounts were still higher than the overall average purchase value.

	customer_id	purchase_amount
▶	1676	90
	1674	62
	1673	73
	1671	73
	1667	64
	1662	86
	1659	66
	1656	81
	1654	93
	1652	80
	1650	63
	1649	69
	1648	78
	1647	77
	1645	90

Total 839 rows returned.

3. Highest-Rated Products

The products with the highest average customer review ratings were identified to understand which products received the strongest customer feedback.

	item_purchased	average_review_rating
▶	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Skirt	3.78

4. Shipping Method Analysis

Average purchase amounts were compared between **Standard** and **Express** shipping customers to determine whether shipping preferences were associated with differences in spending.

	shipping_type	average_purchase_amount
►	Express	60.48
	Standard	58.46

5. Subscribers vs. Non-Subscribers

Customer spending and revenue were compared between subscribers and non-subscribers to evaluate the relationship between subscription status and purchasing behavior.

	subscription_status	total_customer	average_spend	total_revenue
►	Yes	1053	59.49	62645
	No	2847	59.87	170436

6. Discount-Dependent Products

Products with the highest proportion of discounted purchases were identified to determine which products appeared to rely more heavily on promotions.

	item_purchased	discount_rate
►	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

7. Customer Segmentation

Customers were divided into three groups based on their purchase history:

- **New**
- **Returning**
- **Loyal**

This segmentation was used to understand differences in customer engagement and purchasing behavior.

	customer_segment	number_of_customers
►	loyal	3116
	Returning	701
	New	83

8. Top Products Within Each Category

The three most frequently purchased products within each product category were identified to determine category-level product preferences.

	item_rank	category	item_purchased	total_orders
	1	Accessories	Jewelry	171
►	1	Clothing	Blouse	171
	1	Footwear	Sandals	160
	1	Outerwear	Jacket	163
	2	Accessories	Sunglasses	161
	2	Clothing	Pants	171
	2	Footwear	Shoes	150
	2	Outerwear	Coat	161
	3	Accessories	Belt	161
	3	Clothing	Shirt	169
	3	Footwear	Sneakers	145

9. Repeat Purchasers and Subscription Behavior

Customers with more than five previous purchases were analyzed to determine whether frequent buyers were more likely to have an active subscription.

	subscription_status	repeat_buyers
►	Yes	958
	No	2518

10. Revenue Contribution by Age Group

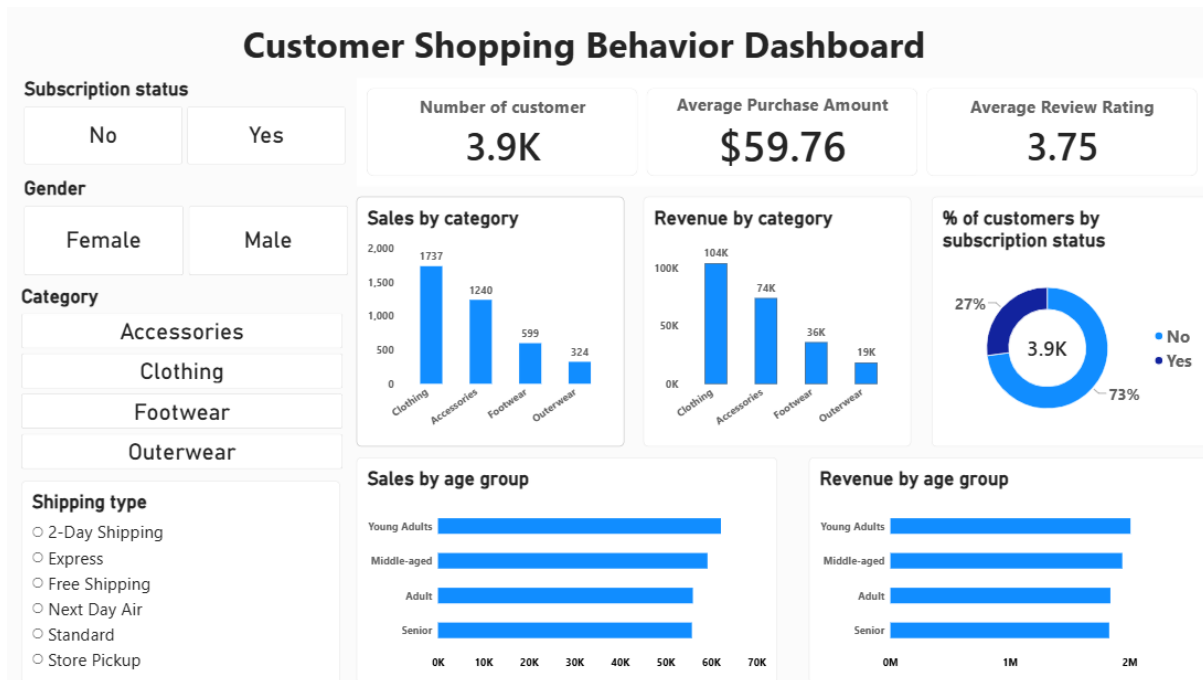
Total revenue was calculated for each age group to identify which customer demographics contributed most to overall revenue.

	age_group	total_revenue
►	Young Adults	62143
	Middle-aged	59197
	Adult	55978
	Senior	55763

5. Dashboard Development in Power BI

The results of the analysis were presented through an **interactive Power BI dashboard**.

The dashboard was designed to provide stakeholders with an easy way to explore customer behavior and identify important patterns across different dimensions.



6. Business Recommendations

Based on the analysis, the following strategies can be considered:

Increase Subscription Adoption

Introduce targeted benefits, exclusive offers, and other incentives to encourage more customers to subscribe.

Strengthen Customer Loyalty

Develop loyalty initiatives that reward repeat customers and encourage returning customers to progress toward the loyal-customer segment.

Optimize Discount Strategies

Review the use of discounts across products to identify where promotions generate value while avoiding unnecessary dependence on price reductions.

Promote High-Performing Products

Use sales performance and customer-rating insights to give greater visibility to products that demonstrate strong customer demand and satisfaction.

Develop Targeted Marketing Campaigns

Focus marketing efforts on customer groups and age segments that contribute significantly to revenue, while tailoring campaigns according to observed shopping behavior.